Annotated Bibliography

Andersen, D. C. (2016). Flow regime effects on mature Populus fremontii (Fremont cottonwood) productivity on two contrasting dryland river floodplains. *The Southwestern Naturalist*, *61*(1), 8-17.

This report analyzes streamflow and water quality by comparing the highly regulated Green River to the free-flowing Yampa. Between 1982 and 2011, Andersen investigates how varying flow regimes influence the productivity of mature Fremont cottonwood trees on these two distinct dryland river floodplains. On a river, streamflow patterns, floodplains, and the water table are affected by dams upstream. The research emphasizes the importance of hydrological conditions, including water availability and flood frequency, on the growth and health of these trees. Using basal-area increment (BAI) to index tree productivity, which is commonly used in studies examining tree response to drought and other hydrological stressors. He also surveyed hydrologic discharge patterns and collected tree cores to assess growth. The study found that BAI was higher in Deer Lodge Park along the Yampa than at Browns Park along the Green, which suggests that the Flaming Gorge Dam has a negative impact on cottonwood trees. The author included many scatter plots and charts to visually represent the data. This report is crucial for understanding the health of the Upper Yampa River Basin, providing essential information for water resource management and conservation efforts. The direct correlation between streamflow and cottonwood productivity opens up an opportunity to adapt the water management framework to maintain healthy populations of cottonwoods in dryland environments. This study is relevant to my project because it displays how cottonwoods are inherently resilient, but they can be influenced by certain factors like streamflow.

Day, N. K. (2021). *Assessment of streamflow and water quality in the Upper Yampa River Basin, Colorado, 1992–2018* (No. 2021-5016). US Geological Survey. USGS

This study used streamflow and water-quality data from selected USGS sites to provide a better understanding of how major factors, including land use, climate change, and geological features, may influence streamflow and water quality. Day examined hydrological data and water quality metrics, including parameters such as temperature, pH, and nutrient concentrations. The findings reveal trends in both streamflow patterns and water quality, highlighting the effects of seasonal variations and land use changes on the river ecosystem. Land use and erosion contribute to suspended sediment loads and high phosphorus and nitrogen concentrations. The geologic makeup of the area is also a major factor that influences the sediment load in the water, including Cretaceous-aged sandstones, shales, and major coal beds. This document displays maps and tables that visually represent data, and which I could potentially use to decide where I want to sample water along the river. Cyanobacteria blooms in Stage Coach Reservoir have been increasingly more abundant in recent years, as a result of land use, hydrology, and climate change. Bloom-forming algae can produce toxins that are harmful to humans and animals, degrade water quality, and decrease water supply. This report provides long-term data, which we can use to see changes over time. It also allows us to compare the samples from the Yampa River to water-quality standards to consider how healthy our stream is. This document references many databases that I can utilize for my project, like the National Land Cover Database. Results found that there has been a downward trend in streamflows and an upward trend in temperature related to climate change and increased water use. These findings support the basis of my project and provide insights into how we can better manage the Yampa River Basin, especially in regard to Stage Coach Reservoir.

Hubler, Shannon, et al. "Improved Thermal Preferences and a Stressor Index Derived from Modeled Stream Temperatures and Regional Taxonomic Standards for Freshwater Macroinvertebrates of the Pacific Northwest, USA." *Ecological Indicators*, vol. 160, Mar. 2024, p. N.PAG. *EBSCOhost*, https://doi.org/10.1016/j.ecolind.2024.111869.

This study aimed to identify the relationship between Macroinvertabrates and temperature because of their ability to measure stream health and their varying sensitivity based on species. This study took samples from 3501 sites across the Pacific Northwest and used the Maximum Weekly Maximum Temperature which indicates the highest thermal stress. Using this dataset they were able to find what Macroinvertabrates were able to survive in the lowest possible temperatures and the highest temperatures. Using three different measurements, they were able to characterize the relationship between different Macroinvertabrates and the corresponding MWMT. They looked at the, 1) preferred temperatures, 2) temperature limits and 3) their reactions to temperature changes. Using this the researchers were able to develop the Macroinvertebrate Thermal Tolerance Index which can be used in rivers and streams around the world for research just like ours for this project!

Noa-Yarasca, E., Babbar-Sebens, M., & Jordan, C. (2023). An improved model of shade-affected stream temperature in Soil & Water Assessment Tool. *Hydrology & Earth System Sciences*, *27*(3), 739–759. https://doi.org/10.5194/hess-27-739-202

This study aims to identify if regrowth efforts in riparian vegetation to reduce stream temperatures are effective. They use the Soil & Water Assessment Tool (SWAT) model to look at the factors shade plays in the overall health of a stream. They integrated the new energy balance equation into SWAT and tested the Dairy McKay watershed in Oregon in four different

parameters. 1) current conditions of riparian area, 2) full restoration, 3) efficient restoration and 4) no vegetation. They found that the current riparian vegetation performed well in keeping streams at survivable temperatures for species but efficient restoration was found to have very similar results but was significantly lower in cost making it more likely to be used in areas where warming temperatures are an issue. This study can help this project greatly in finding different methods to test our observations off of.

Roon, D. A., Dunham, J. B., & Groom, J. D. (2021). Shade, light, and stream temperature responses to riparian thinning in second-growth redwood forests of northern California. *PLoS ONE*, *16*(2), e0246822. https://doi.org/10.1371/journal.pone.0246822

This study highlights the relationship between shade, and stream health in the context of the thinning of red wood forests in California and how this already tragic situation also directly affects aquatic habitats. Observing three different watersheds using Before After Control Impact in the redwood forests of Northern California the scientists found that the thinning of the trees had increased the average temperatures of each stream. These conditions were found to be primarily dominant in the summer but less extreme observations were found in the fall and spring. This study was able to help scientists determine the extent of what deforestation and thinning of riparian areas can do and can help this project to understand why shade is important for impacted water temperatures.